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Off-ridge roof ventilation device

Abstract

An off-ridge roof ventilation device for placing on an opening of a roof deck. The ventilation device comprises an elongated main hood with a curved shape. The right end of the main hood is closed by a right-side panel, and the left end of the main hood is closed by a left-side panel. The main hood comprises a perforated top face and a venting opening on a bottom face. A throat panel is placed on the bottom face of the main hood in a manner to reduce the venting opening. A perforated interior panel is placed between the top face of the main hood and the throat panel to eliminate wind driven rain and/or snow even in high wind situations. The perforations are preferably designed as mini-louvers, wherein the mini-louvers of the main hood and the mini-louvers of the throat panel comprise the same shape and count the same number.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 16/709,153, filed on Dec. 10, 2019, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/864,761, filed on Jun. 21, 2019, which are incorporated by reference herein in their entirety.

FIELD OF THE INVENTION

(1) The present invention relates generally to ventilation devices, and more particularly, to an off-ridge roof ventilation device for placing on an opening of a roof deck.

BACKGROUND OF THE INVENTION

(2) Ventilation is the intentional circulation of outdoor and indoor air into and out of enclosed spaces, to dilute and displace indoor pollutants and increase thermal comfort and dehumidification.

(3) Natural ventilation is the intentional passive circulation of outdoor and indoor air into and out of an enclosed space, such as an inside of a building. The passive circulation occurs by air flowing from the outside into the inside of the building and vice versa through planned openings of a building without moving parts. Natural ventilation relies on diffusion, wind pressure, or the stack effect. The internal heat of a building gains by the heat from people inside the building, resulting in a temperature difference between the interior and exterior of the building. The temperature difference creates a pressure difference between the indoor and outdoor space of the building making the air flow from the outdoor space to the indoor space of the building and vice versa.

(4) A roof is the top covering of a building, supported by the walls of the building. The roof protects the interior of the building against rain, snow, sunlight, extreme temperatures, and wind. Roofs are normally either flat or pitched. The pitch refers to the steepness or slope of the roof. Generally, pitched roofs are configured to redirect or prevent accumulation of water and/or snow. Thus, the pitch is typically greater in areas of high rain and/or snowfall.

(5) State of the art ventilation devices are usually mounted on a roof and have a low profile to minimize damage from high winds. In case of a pitched roof, the ventilation device is often placed away from a ridge of the roof to minimize damage from high winds. Such ventilation devices are known as off-ridge roof ventilation devices.

(6) An off-ridge roof ventilation device is usually positioned over an opening in the roof sheathing leading to an attic or other space of the building. The opening of the roof deck is cut from the roof sheathing, wherein the location of the opening of the roof deck is usually at least 12" off the lower roof edge. The off-ridge roof ventilation device is normally configured to cover the opening of the roof deck completely and seal the opening of the roof deck without blocking airflow.

(7) State of the art off-ridge roof ventilation devices usually comprise a hood with a front opening to allow circulation of outdoor and indoor air through an opening in the roof deck and the front opening. The front opening is usually covered by a screen to prevent small animals or debris from entering the off-ridge roof ventilation device. Such off-ridge roof ventilation devices are disadvantageous in high wind situations, because rain and/or snow can pass through the front opening and reach the interior of the building through the opening in the roof deck.

(8) Accordingly, there is an established need for an off-ridge roof ventilation device, which completely eliminates wind driven rain and/or snow even in high wind situations. The off-ridge

roof ventilation device should preferably also be aerodynamic and prevent small animals and/or debris from entering the ventilation device without a need for a screen.

SUMMARY OF THE INVENTION

(9) The present invention consists of an off-ridge roof ventilation device for placing on an opening of a roof deck, the off-ridge roof ventilation device comprising an outer hood having a perforated top face and a venting opening on a bottom face. The venting opening is intended to face an opening on a roof deck. A throat or water-retaining panel is arranged on the bottom face of the main hood in a manner to reduce the venting opening. A perforated interior panel is arranged between the top face of the main hood and the throat panel, and is configured to allow inside air venting while retaining outside rain and directing the rain towards the throat panel.

(10) The interior panel along with the aerodynamic hood completely eliminates wind driven rain even in high wind situations. No front opening, no screen, and no exterior baffle are needed. The perforations on the lower portion of the hood allow the air to flow through the ventilation device. The ventilation device is maintenance free and totally resistant to wind uplift and wind driven rain. Without the need for having an open front as is typically known in the art, such a structure provides superior resistance to wind driven rain by not allowing wind to enter the ventilation device.

(11) In a first implementation of the invention, the off-ridge roof ventilation device comprises a main hood configured to cover an opening on a roof deck. The main hood is provided with perforations and an open bottom. A throat panel is arranged on the open bottom of the main hood and is configured to partially cover the open bottom of the main hood leaving a venting opening in the open bottom of the main hood, the venting opening configured to be arranged over the opening on the roof deck. A perforated interior panel is arranged between the top face of the main hood and the throat panel.

(12) In a second aspect, the throat panel can extend from a front edge of the main hood towards a rear edge of the main hood, and the venting opening of the main hood can extend from a rear edge of the throat panel to the rear edge of the main hood.

(13) In another aspect, the throat panel can include an interior baffle configured to at least partially block outside air entering through the perforations of the main hood from flowing towards the venting opening of the main.

(14) In another aspect, the interior baffle may extend from the rear edge of the throat panel.

(15) In another aspect, a left side of the main hood may be closed by a left-side panel and a right side of the main hood may be closed by a right-side panel.

(16) In another aspect, each one of the left-side panel and right-side panel can include a face portion and a side flange extending from the face portion.

(17) In yet another aspect, the main hood may include a rear portion and a front portion. The front portion of the main hood can be a solid wall comprising the perforations of the main hood, and the rear portion of the main hood can be a non-perforated solid wall.

(18) In another aspect, the rear portion may be convex and rearward-sloped and the front portion may be convex and frontward-sloped.

(19) In another aspect, the rear portion of the main hood can be reinforced with at least one reinforcement rib.

(20) In another aspect, the perforations in the main hood and/or the throat panel can be formed by a roll forming process.

(21) In yet another aspect, the perforations in the main hood and/or the throat panel can be C-shaped and may include front and rear openings.

(22) In another aspect, the perforations in the main hood and/or the throat panel can be formed as louvers.

(23) In another aspect, the roof ventilation device can further include a front flange extending frontward from a front edge of the main hood, the front flange configured to rest on the roof deck.

(24) In yet another aspect, the front flange can include an upward bent to increase rigidity of the

front flange and roof ventilation device.

(25) In another aspect, the roof ventilation device can include a rear flange extending rearward from a rear side of the main hood, the rear flange configured to rest on the roof deck.

(26) In another aspect, the perforations of the interior panel can be shaped the same as the perforations of the main hood.

(27) In another aspect, the perforations of the interior panel can be oriented towards the bottom side of the main hood.

(28) In yet another aspect, the perforations of the main hood can be oriented outwardly of the main hood.

(29) These and other objects, features, and advantages of the present invention will become more readily apparent from the attached drawings and the detailed description of the preferred embodiments, which follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The preferred embodiments of the invention will hereinafter be described in conjunction with the appended drawings provided to illustrate and not to limit the invention, where like designations denote like elements, and in which:

(2) FIG. 1 presents a top front isometric view of an off-ridge roof ventilation device according to an embodiment of the present invention;

(3) FIG. 2 presents a bottom front isometric view of the off-ridge roof ventilation device of FIG. 1;

(4) FIG. 3 presents an exploded top front view of the off-ridge roof ventilation device of FIG. 1;

(5) FIG. 4 presents a top front isometric view of the off-ridge roof ventilation device of FIG. 1 mounted on a roof deck according to an illustrative application of the present invention;

(6) FIG. 5 presents a cross-sectional side elevation view of the off-ridge roof ventilation device and roof deck installation of FIG. 4, showing the off-ridge roof ventilation device venting air from inside roof; and

(7) FIG. 6 presents a cross-sectional side elevation view of the off-ridge roof ventilation device and roof deck installation of FIG. 4, showing the off-ridge roof ventilation device stopping wind, rain and other external agents outside the roof from entering the roof.

(8) Like reference numerals refer to like parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(9) The following detailed description is merely exemplary in nature and is not intended to limit the described embodiments or the application and uses of the described embodiments. As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.”

Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. For purposes of description herein, the terms “upper”, “lower”, “left”, “rear”, “right”, “front”, “vertical”, “horizontal”, and derivatives thereof shall relate to the invention as oriented in FIG. 1. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

(10) The illustration of FIG. 1 presents a top front isometric view of an off-ridge roof ventilation device **100** according to an embodiment of the present invention. The ventilation device **100** is designed for placing on an opening **420** of a roof deck **400** (FIGS. 5 and 6) and comprises a main hood **102**. The main hood **102** comprises an elongated body with longitudinal, left and right ends and a curved top surface providing an aerodynamic profile, in a manner that the curved main hood **102** provides superior resistance to wind uplift. In some embodiments, such as the present embodiment, the main hood **102** is formed from a single sheet of metal.

(11) The ventilation device **100** further comprises a front flange **104** extending from a front lateral side of the main hood **102**, and a rear flange **106** extending from a rear lateral side of the main hood **102**. In some embodiments, the front flange **104** can be designed as a stiffening rib or bend to increase rigidity. The ventilation device **100** further comprises a left-side panel **108** and a right-side panel **114**. As shown in FIG. 1 and best shown in FIG. 3, the left side panel **108** is attached to the left end of the main hood **102** in a manner that a face portion **110** of the left side panel **108** closes the left end of the main hood **102**; similarly, the right-side panel **114** is attached to the right end of the main hood **102** in a manner that a face portion **116** of the right-side panel **114** closes the right end of the main hood **102**.

(12) With continued reference to FIG. 1, the main hood **102** comprises a rear portion **132** and a front portion **134**. The rear portion **132** is convex and rearward-sloped, while the front portion **134** is convex and frontward-sloped. The front portion **134** is designed as an integral perforated portion with a plurality or array of perforations **136**. The perforations **136** in the front portion **134** can be formed, for instance and without limitation, by a roll forming process. Each perforation **136** is formed as a mini-louver and is dome-shaped or C-shaped, and comprises a curved bridge portion **140** extending over and spaced-apart from a perforation opening **142** and providing a small front opening **144** and a small rear opening **146**.

(13) As way of example, the openings of the perforations **136** of the front portion **134** of the hood **102** may have a size of 0.42 inches by 0.12 inches. This size leads to openings with 0.04 square inches. Therefore, the front portion **134** of the hood **102** allows a free air space of 0.04 square inches per opening. Thus, a four-foot ventilation device **100** with 3,026 perforations **136** may provide a total of 121 square inches of free air space.

(14) As best shown in FIG. 1, the rear portion **132** of the main hood **102** is substantially solid and can be reinforced with reinforcement ribs **130**. In some embodiments, the reinforcement ribs **130** can be spaced 12" apart from each other and run the entire height of the rear portion **132** of the main hood **102**.

(15) In some embodiments, the left-side panel **108** can be attached to the left end of the main hood **102** of the ventilation device **100** using a Pittsburgh seam. Similarly, the right-side panel **114** can be attached to the left end of the main hood **102** of the ventilation device **100** using a Pittsburgh seam. A ¼" side flange of the left- and right-side panel **108**, **114** can be inserted into the pocket of the Pittsburgh seam. The pocket of the Pittsburgh seam may be filled with a non-hardening seam sealant. The filling of the sealant may be completed prior to the folding of the metal or material used for the main hood **102**. The outer edge of the Pittsburgh seam may then be hammered over the left- and right-side panel **108**, **114**.

(16) The illustration of FIG. 2 presents a bottom front isometric view of the off-ridge roof ventilation device **100** of FIG. 1. According to this illustration, the ventilation device **100** of FIG. 1 also comprises a throat panel **200** and a venting opening **202** arranged on a bottom side of the ventilation device **100** and configured to face the roof opening for purposes that will be hereinafter described. As shown, the throat panel **200** is arranged frontward and parallel to the venting opening **202**, and both the throat panel **200** and the venting opening **202** extend across the hood **202**, from the left-side panel **108** to the right-side panel **114**. The left- and right-side panels **108**, **114** can be attached to the throat panel **200** by a clinch lock attachment, for instance and without limitation.

(17) The illustration of FIG. 3 presents an exploded top front view of the off-ridge roof ventilation

device **100** of FIG. **1**. As shown, the left-side panel **108** includes the aforementioned face portion **110** and a side flange **112** extending from the face portion **110** of the left-side panel **108**. Furthermore, the right-side panel **114** includes the aforementioned face portion **116** and a side flange **118** extending from the face portion **116** of the right-side panel **114**. The ventilation device **100** further comprises an interior panel **300** with perforations **136**. As better shown in FIGS. **5** and **6**, the interior panel **300** is arranged beneath the hood **102**, and more specifically, between the hood **102** and the throat panel **200** and preferably frontward-sloped. The perforations **136** of the interior panel **300** can have the same shape and size as the perforations **136** of the hood **102**, with the bridge portions **140** of the perforations **136** of the interior panel **300** oriented downward instead of upward.

(18) The illustration of FIG. **4** presents a top front isometric view of the off-ridge roof ventilation device **100** of FIG. **1** mounted on a roof deck **400** according to an illustrative application of the present invention. In the present illustrative application, the roof deck **400** is covered by shingles **410**. The ventilation device **100** can be installed on the roof deck **400** such that the front and rear flanges **104**, **106** extend substantially parallel to the ridge (not shown, but otherwise parallel to the roof shingles **410**), with the rear flange **106** more proximate to the ridge than the front flange **104**. The left- and right-side panels **108**, **114** extend substantially perpendicular to the ridge.

(19) As shown in FIG. **4** and also in FIGS. **5** and **6**, the rear flange **106** can be covered by shingles **410** for a better attachment of the ventilation device **100** on the roof deck **400**. In turn, the side flanges **112**, **118** of the left- and right-side panels **108**, **114** and the front flange **104** can be placed above or adjacent to the shingles **410**, i.e. not covered by the shingles **410**.

(20) The side flanges **112**, **118** of the left- and right-side panels **108**, **114** and the main hood **102** provide a sealing of the opening of the roof deck **400** by the ventilation device **100** thereby contributing to comply with best roofing industry practices.

(21) The illustration of FIG. **5** presents a cross-sectional side elevation view of the off-ridge roof ventilation device **100** and roof deck **400** installation of FIG. **4**. According to this illustration, the ventilation device **100** is placed on the roof deck **400** and completely covers the opening **420** of the roof deck **400**. The opening **420** of the roof deck **400** faces and is in fluid communication with the venting opening **202** of the ventilation device **100**. An underside of the throat panel **200** touches the roof deck **400** close to the opening **420** of the roof deck **400** in a large area, giving the ventilation device **100** a strong attachment to the roof deck **400**. The air is vented from an interior of a building through the off-ridge roof ventilation device **100**, which provides a single piece hood that is aerodynamic in nature with no large openings on the outer surface of the ventilation device **100**.

(22) Additionally, as with reference to FIGS. **5** and **3**, an interior ridge or baffle **204** can be attached to or extend from the throat panel **200** at its top or rear edge and extend inwardly into the hood **102**, such as at a right angle with the throat panel **200**. Preferably, the interior baffle **204** has a low profile or height to allow a relatively large air space between the interior baffle **204** and the hood **102**. For example, in the depicted embodiment, a full 2.5 inches of free air space can be provided between the interior baffle **204** and the hood **102**. With further reference to FIG. **5**, venting air **500** flows from the inside of the building to the opening **420** of the roof deck **402** and from there to the venting opening **202** of the venting device **100**. Once the air is inside the venting device **100**, it is sucked out of the venting device **100** through the perforations **136** of the interior panel **300** and the perforations **136** of the front portion **134** of the main hood **102**, and more specifically, through the perforation opening **142**, front opening **144** and rear opening **146** of each perforation **136**.

(23) In some embodiments, as in the present embodiment, the interior panel **300** can comprise the same number of perforations **136** as on the front portion **134** of the hood **102**. As mentioned heretofore, the perforations **136** of the interior panel **300** can be shaped the same as the perforations **136** of the front portion **134** of the main hood **102**. However, the perforations **136** of the interior panel **300** can be oriented inwardly to the interior of the venting device **100**. In contrast, the perforations **136** of the front portion **134** of the main hood **102** can be oriented outwardly of the

interior of the venting device **100**.

(24) The illustration of FIG. **6** presents a cross-sectional side elevation view of the off- ridge roof ventilation device **100** and roof deck **400** installation of FIG. **4**, in a situation in which outside wind **600** is impacting the installation. The wind **600** can be, for instance, a high-speed wind that strikes frontally on the front portion **134** of the ventilation device **100**. The wind **600** can drive rain and/or snow to the front portion **134** of the ventilation device **100**. In case the air, rain and/or snow is driven through the openings of the perforations **136** of the front portion **134** of the ventilation device **100**, the rain and/or snow will fall on the interior panel **300**. The air, rain and/or snow is then further mitigated and/or retained by the interior panel **300**; for example, the rain and/or snow flows along the upper, sloped surface of the interior panel **300** to a lower edge of the interior panel **300** and drops on throat panel **200** through a gap **310** which is formed between the lower edge of the interior panel **300** and the throat panel **200**. From there the rain and/or snow can flow out of the ventilation device **100** through the first three rows of the perforations **136** of the front portion **134**, and any remaining humidity inside the hood **102** can eventually evaporate. The baffle **204** at the top end of the throat panel **200** contributes to retain any air and/or water within the throat panel **200** and prevent it from passing through the venting opening **202** of the ventilation device **100** and the opening **420** of the roof deck **400**.

(25) Since many modifications, variations, and changes in detail can be made to the described preferred embodiments of the invention, it is intended that all matters in the foregoing description and shown in the accompanying drawings be interpreted as illustrative and not in a limiting sense. Furthermore, it is understood that any of the features presented in the embodiments may be integrated into any of the other embodiments unless explicitly stated otherwise. The scope of the invention should be determined by the appended claims and their legal equivalents.

Claims

1. An off-ridge roof ventilation device for placing on an opening of a roof deck, comprising: a main hood configured to cover an opening on a roof deck, the main hood comprising perforations and an open bottom; a throat panel arranged on the open bottom of the main hood and configured to partially cover the open bottom of the main hood leaving a venting opening in the open bottom of the main hood, the venting opening configured to be arranged over the opening on the roof deck; and an interior panel arranged between the main hood and the throat panel, the interior panel separate from the throat panel and comprising: perforations extending therethrough; a first end attached to and extending from the main hood; and a second end, spaced apart from the main hood and overlying the throat panel.
2. The roof ventilation device of claim 1, wherein the throat panel extends from a front edge of the main hood towards a rear edge of the main hood, and the venting opening of the main hood extends from a rear edge of the throat panel to the rear edge of the main hood.
3. The roof ventilation device of claim 2, wherein the throat panel comprises an interior baffle configured to at least partially block outside air entering through the perforations of the main hood from flowing towards the venting opening of the main hood.
4. The roof ventilation device of claim 3, wherein the interior baffle extends from the rear edge of the throat panel.
5. The roof ventilation device of claim 2, wherein a left side of the main hood is closed by a left-side panel and a right side of the main hood is closed by a right-side panel.
6. The roof ventilation device of claim 5, wherein each one of the left-side panel and right-side panel comprises a face portion and a side flange extending from the face portion.
7. The roof ventilation device of claim 1, wherein the main hood comprises a rear portion and a front portion, wherein the front portion of the main hood is a solid wall comprising the perforations of the main hood, and the rear portion of the main hood is a non-perforated solid wall.

8. The roof ventilation device of claim 7, wherein the rear portion is convex and rearward-sloped and the front portion is convex and frontward-sloped.
 9. The roof ventilation device of claim 7, wherein the rear portion of the main hood is reinforced with at least one reinforcement rib.
 10. The roof ventilation device of claim 1, wherein the perforations in at least one of the main hood and the throat panel are formed by a roll forming process.
 11. The roof ventilation device of claim 1, wherein the perforations in at least one of the main hood and the throat panel are C-shaped and comprise front and rear openings.
 12. The roof ventilation device of claim 1, wherein the perforations in at least one of the main hood and the throat panel are formed as louvers.
 13. The roof ventilation device of claim 1, further comprising a front flange extending frontward from a front edge of the main hood, the front flange configured to rest on the roof deck.
 14. The roof ventilation device of claim 13, wherein the front flange comprises an upward bend.
 15. The roof ventilation device of claim 1, further comprising a rear flange extending rearward from a rear side of the main hood, the rear flange configured to rest on the roof deck.
 16. The roof ventilation device of claim 1, wherein the perforations of the interior panel are shaped the same as the perforations of the main hood.
 17. The roof ventilation device of claim 1, wherein the perforations of the interior panel are formed as louvers oriented towards the open bottom of the main hood.
 18. The roof ventilation device of claim 1, wherein the perforations of the main hood are formed as louvers oriented outwardly of the main hood.
 19. An off-ridge roof ventilation device for placing on an opening of a roof deck, comprising: a main hood configured to cover an opening on a roof deck, the main hood comprising perforations and an open bottom, wherein a left side of the main hood is closed by a left-side panel and a right side of the main hood is closed by a right-side panel; a throat panel arranged on the open bottom of the main hood and extending from a front edge of the main hood towards a rear edge of the main hood, wherein the throat panel is configured to partially cover the open bottom of the main hood leaving a venting opening in the open bottom of the main hood, wherein the venting opening extends from a rear edge of the throat panel to the rear edge of the main hood and is configured to be arranged over the opening on the roof deck; and an interior panel arranged between the main hood and the throat panel, the interior panel separate from the throat panel and comprising: perforations extending therethrough; a first end attached to and extending from the main hood; and a second end, spaced apart from the main hood and overlying the throat panel.
 20. An off-ridge roof ventilation device for placing on an opening of a roof deck, comprising: a main hood configured to cover an opening on a roof deck, the main hood comprising a rear portion and a front portion, wherein the front portion of the main hood is a convex, frontward-sloped solid wall provided with perforations, and the rear portion is a convex, rearward-sloped non-perforated solid wall, wherein a left side of the main hood is closed by a left-side panel and a right side of the main hood is closed by a right-side panel; a throat panel arranged on the open bottom of the main hood and extending from a front edge of the main hood towards a rear edge of the main hood, wherein the throat panel is configured to partially cover the open bottom of the main hood leaving a venting opening in the open bottom of the main hood, wherein the venting opening extends from a rear edge of the throat panel to the rear edge of the main hood and is configured to be arranged over the opening on the roof deck; and an interior panel arranged between the main hood and the throat panel, the interior panel separate from the throat panel and comprising: perforations extending therethrough; a first end attached to and extending from the main hood; and a second end, spaced apart from the main hood and overlying the throat panel; wherein the perforations in the main hood and the throat panel are formed as louvers.
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